Conformal gravity must choose: exceptional-point kinematics, vacuum decay, and the sample space of Weyl gravity

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July 12, 2026

Abstract

Mannheim's conformal (Weyl) gravity is the flagship application of the $\mathcal{PT}$/antilinearity program, and its quantum-mechanical prototype is the Pais–Uhlenbeck oscillator at *equal frequencies*: the $1/k^4$ propagator is, mode by mode, the degenerate limit of two opposite-residue poles—a Jordan block. We analyze the theory in the stochastic currency of the preceding papers of this series, with four results. (i) *Kinematics*: every linearized spin-2 field mode ($k > 0$) of conformal gravity sits exactly at an exceptional point of the similarity dictionary, whose per-mode cost we lower-bound for every normal-target (mode-built) dictionary (up to an $O(1)$ factor, the EP-driven cond $\to 2/|\delta|$ at $\omega = 1$, following the relative detuning δ/ω along a dispersion; massive quadratic gravity approaches the exceptional line in the ultraviolet with EP-driven cost growing as k^2), while the Hermitian ghost realization passes through the point exactly unitarily. (ii) *A no-representation lemma*: a real-eigenvalue Jordan-block Hamiltonian (finitely many configurations) admits *no* closed stochastic-process representation under any similarity whatsoever—the obstruction at the conformal point is structural, not a diverging cost—and the lemma lifts to the free field level: the M-quantum sector of a conformal mode carries a single Jordan block of size $M+1$ (secular growth t^M, the constructive form of Smilga's infinite-order exceptional point), and no metric that is bounded with bounded inverse in the occupation-number Fock norm yields a stochastic representation on a conformal mode's Fock tower. This, we argue, recasts the published Bender–Mannheim/Smilga dispute: "unitarity is broken" is the closed-currency statement, while unitary evolution via nonstationary states we read, in this currency, as the survival-conditioned bookkeeping of the dilation—an open-system reading. The dipole ghost's open process is a one-way cascade whose indivisibility enters, at the mechanical cell, through conditional mass creation—a mechanism new to the taxonomy (the higher Fock sectors add ordinary divisor negativity—a convention-independent sign; the mass-creation reading is stated relative to the minimal survival-conditioning convention, whose definition and justification the text makes explicit). (iii) *Vacuum decay*: at the scale-invariant ($\delta = 0$) point the ghost sector's golden-rule rate sits on the contact (s-flat) rung of the dimension ladder, $\Gamma/V \sim \Lambda^4/\alpha_g^2$—a maximally generous lower bound, since consistently regulating the split poles ($\delta > 0$) only raises it (for modes of relative detuning $\ll 1$)—with a dimensionless coupling in place of the Planck suppression; pricing the Lorentz-violating-cutoff rescue then forks on α_g—the face-value rotation-curve matching ($|\alpha_g| \sim 3 \times 10^{109}$) makes the bound vacuous but that branch fails solar-system tests on published (Mannheim-disputed) analyses, while an unfixed α_g within eight decades of unity (a conservative figure) forces the cutoff to macroscopic wavelengths; the larger-coupling ray is closed on the face-value branch by

point-collapse—every leading-order gravitational (one-exchange) tree observable scales as $1/\alpha_g$, so the fit pins α_g to a single, solar-system-dead value—and on Mannheim's branch by an explicit naturalness prior, stated as such. (iv) *The trilemma closes*: coupled ghosts, empirical Lorentz invariance, definite configurations—pick two (at tree level). Conformal gravity's no-ghost quantization has already picked: at the exceptional point its Jordan dynamics admits no *closed* definite-configurations seat within the stated dictionary classes—a free/linearized-level statement—so that program is born on the exit-the-sample-space horn, while the Hermitian ghost realization keeps configurations at precisely horn 1's price; we price both seats. The *adjudicated* escape routes map onto the horns (one horn-adjacent, one premise-disputing; the recent 2024–2026 entries, the older alternative quantizations, and the Bohmian route remain pending).

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every theorem is stated with a full proof, and every numerical claim is reproducible from the public scripts or, where stated, from the in-text analytic computations (see Code availability); the originating conversation, project briefing, and full research notebook are preserved in the program's hub repository, `https://github.com/embeddetech/ghosts-program`.

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1 Introduction

Fourth-order gravity buys renormalizability with a ghost [36], and every school of quadratic gravity has a prescription for living with it: Mannheim's conformal program quantizes the ghost away with $\mathcal{PT}$ methods and zero-norm states [7, 8, 10]; Donoghue and Menezes keep it and pay in microcausality [39]; Anselmi's fakeons remove it by fiat at the level of the propagator prescription [40]; Salvio and Strumia keep positive energies at the price of a negative-norm configuration space [35]; and the agravity analysis of Gross–Strumia–Teresi–Zirilli, asking whether negative kinetic energy is even metastable, answers that for gravitationally coupled ghosts the runaway is "cosmologically slow"—a Planck-suppression statement—while conceding that the field-theoretic rate is "naively infinite" and that they "do not know the true rate" [32].

The preceding papers of this series developed a currency in which such prescriptions can be compared [1, 2, 3]: quantum theory as indivisible stochastic processes on a configuration space (Barandes [4, 5]), in which ghosts are a recurrence-classification problem (conservative → transient → explosive) [1], the divergent ghost-vacuum rate of field theory is an instantaneous-state pathology whose resolution is the choice of sample space [2], and the whole subject obeys a trilemma: *coupled ghosts, empirical Lorentz invariance, definite configurations—any two, never all three* (at tree level) [2].

This paper aims that currency at the sharpest target: pure conformal (Weyl) gravity, whose ghost is not a separate massive pole but a *degeneracy*—the $1/k^4$ propagator is, in Mannheim's own presentation, the limit of two poles of opposite residue as their masses merge [9], and the corresponding mechanical prototype, the equal-frequency Pais–Uhlenbeck (PU) oscillator, is a Jordan-block Hamiltonian: nondiagonalizable, and—by Bender and Mannheim's own theorem—possessing *no Hermitian counterpart* [8]. Whether that theory is unitary is disputed in print: Bender and Mannheim construct unitary evolution on nonstationary states [8]; Smilga concludes that for this nonstandard realization "unitarity is broken in the equal frequencies limit" [14],

later recasting the point as an exceptional point of infinite order [15]; alternative quantizations trade the difficulty for other costs [31, 42]; and the quantization remains contested into this year [16, 17, 18, 19]. Two of the newest entries sit closest to this paper's machinery and deserve their coordinates on our map: the ghost-free quantization via non-unitary similarity transformations [17] is a similarity dictionary in precisely the sense of Sec. 3, so at the equal-frequency point it meets Lemma 4.1 sector-wise (any sector-preserving Dyson map), or Theorem 4.5 if its Dyson map is bounded with bounded inverse in the occupation-number Fock norm (an unbounded, sector-mixing Dyson map falls outside both, into the open class of Claim 6.1); and the Bohmian analysis of the degenerate cell [19] equips the same kinematics with *deterministic* configuration trajectories—a different beable ontology, whose inequivalence findings concern trajectory dynamics and are complementary to, not in tension with, the stochastic-representation lemma proved here. Beyond the PU quantization dispute, Holdom argues (in 0+1d models aimed at quadratic gravity) that ghost theories are intrinsically unitary and perturbatively stable once an appropriate inner product defines the Born rule [20]—an inner-product construction this paper's currency would price as a sample-space exit—the $\mathcal{PT}$ route has been carried to the full quadratic theory [21], a Krein-space quantization of a four-derivative QFT with a generalized Born rule has appeared as this manuscript was completed [22], a direct-sum quantization recasting the massive tensor ghost as an inverted-harmonic-oscillator instability held off the asymptotic spectrum has been carried to the full 4d theory [23], the massive ghost has been argued—conditionally on a ghost–Faddeev–Popov bound state—to confine into zero-norm BRST-quartet bound states [30], complex-mass ghost pairs have been argued to bind into states satisfying reflection positivity [24], an LSZ scattering construction has been built atop negative-norm-projecting quantizations of the quadratic theory [25], and ghost fields have been argued never to appear as isolated asymptotic particles at all [26]; all of these enter the trilemma registry as pending classifications rather than adjudicated rows.

Our results, in brief. Section 3: the exceptional point is not a corner of parameter space but the entirety of conformal gravity's spin-2 kinematics—every mode ($k > 0$) of that sector sits at it, and we measure what that costs any similarity dictionary with a normal target built on the mode decomposition. Section 4: a short lemma shows a real-eigenvalue Jordan-block Hamiltonian (finitely many configurations) admits no closed stochastic-process representation under any similarity; the open (dilated) representation exists and is a one-way conditioned cascade with a new indivisibility mechanism. The Bender–Mannheim/Smilga dispute dissolves into two currencies. Section 5: the vacuum-decay rate of the conformal ghost sector sits on the contact rung of the series' dimension ladder, Λ^4/α_g^2 (a maximally generous lower bound—regulating the split poles only raises it for modes of relative detuning $\ll 1$), and pricing the Lorentz-violating-cutoff rescue forks on the value of α_g—with neither branch tenable (each on a stated closure—observational death or a naturalness prior). Section 6: the trilemma closes, with every adjudicated escape route priced against a horn—one route (Merlin) horn-adjacent, one (metastability) disputing the rate premise rather than picking a horn—and several recent entries logged as pending.

2 The currency, in one page

We use the stochastic–quantum correspondence in the form of [4, 5, 1]: configurations carry the ontology; dynamics is a family of (generally indivisible) stochastic maps $\Gamma(t)$. A *closed representation* in a fixed beable basis is a family $\Gamma(t) = |V(t)|^2$ entrywise with $V(t) = S\mathrm{e}^{-\mathrm{i}Ht}S^{-1}$ for a fixed invertible S (the beable basis is chosen once, so S carries no time dependence), with

$\Gamma(t)$ column-stochastic for all t; Γ is *unistochastic* when $V(t)$ is unitary—the case a positive metric realizes, which is sufficient though not necessary for stochasticity of $|V|^2$. Unbounded or lossy systems get the *minimal*, generally sub-stochastic (probability-losing) process, with unitary dilations as the conservative completions [1]. Three prior results are load-bearing here. (a) For diagonalizable H with real spectrum, the closed representation exists exactly when a positive metric intertwines H with a Hermitian partner (the converse—that a closed representation forces such a metric—is [1]'s); its cost is the beable rotation and metric conditioning [1]. (b) At an exceptional point of a $\mathcal{PT}$ dimer the metric collapses with Petermann scaling while the *open* description passes through smoothly: the EP is a coordinate singularity of the dictionary, not of the process [1]. (c) The trilemma of [2]: a coupled ghost sector, empirical Lorentz invariance, and definite configurations cannot coexist (at tree level); each pair names a known physics program.

3 Kinematics: conformal gravity's spin-2 sector lives at the exceptional point

The quadratic (TT-sector) action of conformal gravity, $I_W = -\alpha_g \int d^4x\sqrt{-g}\, C_{\lambda\mu\nu\kappa}C^{\lambda\mu\nu\kappa}$ with α_g dimensionless, linearizes to $I_W^{(2)} = -\frac{\alpha_g}{2}\int(\Box K_{\mu\nu})^2$ ($K_{\mu\nu}$ the traceless part of the metric perturbation), giving a propagator $\propto P_2/(\alpha_g k^4)$ (P_2 the transverse-traceless spin-2 projector; the explicit tensor normalization is our convention, and the overall constant is fixed below, where the flat-space exchange is written $1/(2\alpha_g k^4)$; [9] states the $1/k^4$ form). Mannheim presents the $1/k^4$ explicitly as the degenerate limit of two opposite-residue $1/k^2$ poles, and the equal-mass limit as a nondiagonalizable Jordan-block Hamiltonian [9, 10], whose characteristic zero-norm states Bender and Mannheim exhibit explicitly [8]. The linearized particle content—massless spin-2, spin-1, and the ghost partner—was catalogued by Riegert [45]; the exceptional-point statements of this paper concern the spin-2 sector, where the degeneracy lives. Mechanically: each spatial mode k is a PU oscillator [6] with frequencies $\omega_1(k) = \omega_2(k) = k$—the equal-frequency point, identically in k ($k > 0$).

What does the degeneracy cost? The first-order mode matrix $A(\omega_1, \omega_2)$ of the PU oscillator (variables $q_1 = x$, $q_2 = \dot{x}$, p_1, p_2) has eigenvalues $\pm i\omega_1, \pm i\omega_2$ merging at $\delta = \omega_1 - \omega_2 = 0$ into two 2×2 Jordan blocks. Numerically (`scripts/t4_ep_dictionary.py`): the eigenvector-matrix condition number—whose divergence is set by the scaling-invariant angle between the merging eigenvectors, which no column rescaling can repair, so that up to an $O(1)$ factor it lower-bounds the conditioning of *any* similarity dictionary with a normal target built on the mode decomposition (normal modes, isospectral partner, quadratic-level Bender–Mannheim map—each has conditioning at least the merging pair's spectral-projector norm, itself $\sim 1/\text{angle}$; a normal target is unitarily diagonalizable and the unitary factor leaves the condition number unchanged, reducing the bound to this diagonal case; this applies for $\delta \neq 0$, where a normal target exists—at $\delta = 0$ there is none, and Lemma 4.1 supersedes)—diverges as

$$\text{cond}(V) \;=\; \frac{2.00}{|\delta|} \qquad \text{(fitted power } -0.993 \text{ over five decades)}, \tag{1}$$

(here V denotes the eigenvector matrix of the mode matrix A, distinct from the dictionary evolution $V(t) = Se^{-iHt}S^{-1}$ of Sec. 2) with the far-from-EP control at cond = 6.8 and numerical defectiveness (rank loss to $\sqrt{\text{machine precision}}$, as expected for a defective pair) at $\delta = 0$. The -1 power is derived, not merely fitted: the frequencies are exact eigenvalues of the family, so the detuning is a *non-generic* unfolding direction of the exceptional point—the eigenvalue

splitting is linear in δ and the eigenvector angle within each merging pair is $\propto \delta$, rather than the generic $\sqrt{\delta}$ of a random perturbation. The angle's coefficient is derived, not only measured: the mode matrix's eigenvector for eigenvalue λ is the chain vector $(1, \lambda, \beta\omega_1^2\omega_2^2/\lambda, \beta\lambda^2)$ (β a fixed kinetic normalization), and linearizing the merging pair of the detuned matrix at $\beta = \omega = 1$ gives eigenvector difference $\delta\,(0, \mathrm{i}, \mathrm{i}, -2)$ against $v = (1, \mathrm{i}, -\mathrm{i}, -1)$, hence pair angle $|\delta|\sqrt{6-1}/2 = (\sqrt{5}/2)\,|\delta|$ (here $6 = \|(0, \mathrm{i}, \mathrm{i}, -2)\|^2$ and the subtracted 1 is the squared component along v, leaving the perpendicular part; script-emitted: angle$/|\delta| = 1.11802$ at $\delta = 10^{-4}$). The two coefficients are one law at this $\beta = \omega = 1$ normalization (the convention-independent content is the $1/|\delta|$ divergence), linked by the script-emitted identity $\mathrm{cond}(V) \times \mathrm{angle} \to \sqrt{5}$ (computed 2.2363 against $\sqrt{5} = 2.2361$): $\mathrm{cond}(V) = \sqrt{5}/\mathrm{angle} = 2/|\delta|$ (the law is even in the signed detuning). The per-mode law (1) is stated at fixed $\omega = 1$, where absolute and relative detuning coincide; along a dispersion the EP-driven part of the cost follows the dimensionless *relative* detuning δ/ω, which is what the units-only control below isolates. Along the Stelle-type dispersion of *massive* quadratic gravity ($\omega_1 = k$, $\omega_2 = \sqrt{k^2 + m_2^2}$), the detuning magnitude $|\delta(k)| \simeq m_2^2/2k$ vanishes in the ultraviolet and the measured cost grows as $k^{5.0}$, which a units-only control ($\omega_1 = 2k$: $k^{3.0}$) decomposes into a variable-normalization factor k^3 times an *EP-driven excess* $k^{2.0}$—exactly the inverse relative detuning $(|\delta|/k)^{-1} \sim (k/m_2)^2$. Massive quadratic gravity thus approaches the exceptional line mode-by-mode in the UV; conformal gravity, $m_2 = 0$, sits on it at every k.

A one-loop check sharpens the scope of this statement. The ghost pole acquires a width—$\mu^2 = m_r^2 + \mathrm{i}\gamma$ with $\gamma = \xi^2 m_r^2 N_{\mathrm{eff}}/(320\pi)$ (m_r the real pole mass, ξ the ghost–matter coupling, N_{eff} the effective loop multiplicity) [39]—and one might expect the width to detune the theory off the exceptional line. It does not: the shift is Lorentz-invariant in q^2, so at spatial momentum k the frequency-space width $\gamma/2E_k$ time-dilates away at exactly the $1/k$ rate of the real detuning magnitude $m_2^2/2k$; both components of the complex detuning, of magnitude $|m_2^2 + \mathrm{i}\gamma|/2k$, vanish together, and the cost still diverges, suppressed only by the k-independent factor $[1+(\gamma/m_2^2)^2]^{-1/2}$ ($\approx 1 - 2 \times 10^{-4}$ at $\xi = 1$, $N_{\mathrm{eff}} \approx 20$, with $m_r \simeq m_2$ near the exceptional line). What the loop does change is the approach: the detuning ray is tilted into the complex plane by the fixed angle $\arctan(\gamma/m_2^2)$, and the coalescing pair is (real, complex) at every finite k—a resonance-type exceptional point with unequal linewidths, not a real-axis $\mathcal{PT}$-symmetric one. We flag explicitly that the saturation a fixed-linewidth (cavity-photonics) intuition would predict, $\mathrm{cond}_{\max} \sim 2\omega/\Gamma_\mu = 640\pi/(\xi^2 N_{\mathrm{eff}})$ (with $\Gamma_\mu \equiv \gamma/m_r$ the frozen pole width as a frequency), is an artifact of freezing the width in frequency; the relativistic dispersion removes it. Scope: this is the one-loop, weak-coupling ($m_2 \ll M_{\mathrm{Pl}}$) pole statement; off-shell mode matrices at $|q^2| \gg m_2^2$ remain tree-level claims.

Meanwhile the Hermitian ghost realization—the ghost-spectrum theory on real configurations $(x, \dot{x})$, a *different* operator not similar to the confluent Jordan one—passes through the exceptional point exactly: on a grid at $\omega_1 = \omega_2$, $\left\|H - H^\dagger\right\| = 8 \times 10^{-15}$ and $\Gamma = |U|^2$ is doubly stochastic to 1.1×10^{-14} (re-verifying [1] at the field-kinematics framing). The singularity is the dictionary's alone. The dictionary, however, is what the no-ghost program *is*; the next section shows the failure at $\delta = 0$ is not a divergence but a nonexistence.

4 No closed sample space at the conformal point

Lemma 4.1 (no closed process at a Jordan point)**.** *Let H be a matrix on finitely many configurations with a Jordan block of size $m \geq 2$ at a real eigenvalue (hence nondiagonalizable).*

Then there is no invertible S for which $\Gamma(t) = |S\mathrm{e}^{-\mathrm{i}Ht}S^{-1}|^2$ is column-stochastic—in particular, unistochastic or doubly stochastic—for all t: no metric, no partner, no beable rotation yields a closed stochastic representation of any kind.

Proof. If $\Gamma(t) = |V(t)|^2$ is column-stochastic on n configurations, then

$$\sum_{ij} |V_{ij}(t)|^2 = \sum_j \big(j\text{th column sum of } \Gamma\big) = \sum_j 1 = n,$$

so $\|V(t)\|_2 \le \|V(t)\|_F = \sqrt{n}$ is bounded in t. But $\left\|\mathrm{e}^{-\mathrm{i}Ht}\right\| \gtrsim t^{\,m-1}$ is unbounded (real eigenvalue $\Rightarrow |\mathrm{e}^{-\mathrm{i}\lambda t}| = 1$, so the Jordan nilpotent part is undamped), and $\|V(t)\| \ge \left\|\mathrm{e}^{-\mathrm{i}Ht}\right\| / \mathrm{cond}(S)$ is unbounded for every invertible S. □

Remark 4.2. The correspondence itself requires only per-time unistochasticity [1]—a weaker property than $V(t)$ being unitary, since unistochasticity fixes only the moduli of some unitary, not of V. The lemma is therefore stated for the weakest relevant condition (column-stochasticity of $|V|^2$) and excludes the unistochastic representations a fortiori. The core mechanism—column-stochasticity pinning $\|V(t)\|_F = \sqrt{n}$ against the unbounded Jordan growth of $\left\|\mathrm{e}^{-\mathrm{i}Ht}\right\|$ under any fixed similarity—appears for the size-2 boundary case ($s = s_*$) in [1]; the lemma extends that exclusion to arbitrary block size m, to any real eigenvalue, and, below (Theorem 4.5), to the Fock tower.

Numerically (`scripts/t4_jordan_process.py`): growth exponents 1.000, 1.999, 2.997 for $m = 2, 3, 4$ (with prefactors converging to $1/(m-1)!$); the similarity inequality verified with margin; the detuned control saturates at $\|U\| \sim 2/\delta$ as diagonalizability requires. (A numerical remark worth recording: machine-epsilon perturbations split a Jordan block by $\sim \epsilon^{1/m}$ and convert polynomial to exponential growth for $t \gtrsim \epsilon^{-1/m}$; the fragility is itself the exceptional-point physics.)

Lemma 4.1 upgrades the kinematics of Sec. 3: away from the conformal point the dictionary exists at diverging cost (1); at it, the dictionary does not exist. Since Bender and Mannheim themselves established that the equal-frequency Hamiltonian "has no Hermitian counterpart" [8]—their diagnosis of exactly the nondiagonalizability on which the lemma bites—the lemma's content in context is the missing corollary: *for the Jordan dynamics, whose lack of a Hermitian counterpart is its nondiagonalizability, there is no closed probability representation on any beables*—under any similarity at the (finite-dimensional) quadratic level, sector by sector on the tower, and, by the tower theorem below, under any bounded dictionary with bounded inverse on the Fock space. The sector-level exclusion carries no boundedness hypothesis—each M-quantum sector, like the mechanical mode matrix itself, is finite-dimensional—so it already covers the quadratic-level avatars of the realized dictionaries and every sector-preserving field dictionary; the bounded-inverse hypothesis of Theorem 4.5 buys the further generality of bounded field dictionaries that mix sectors. For the no-ghost quantization, the option the trilemma calls "definite configurations" is thus, within the stated dictionary classes, structurally absent at the conformal point, not merely expensive—while the Hermitian ghost realization of Sec. 3, a *different* operator not similar to the Jordan one, keeps definite configurations together with the ghost spectrum, which is exactly the seat Sec. 5 prices.

4.1 From the mechanical prototype to the Fock tower

The lemma so far concerns one mode. Second quantization makes it a field statement, constructively.

Proposition 4.3 (sector tower)**.** *Let the one-particle matrix of a mode be a size-2 Jordan pair, $A = \omega\mathbf{1} + N$ with $\omega \in \mathbb{R}$, $Nv = u$, $Nu = 0$ (the equal-frequency PU cell). Then $H = d\Gamma(A)$ preserves each M-quantum sector (dimension $M{+}1$), and on it carries a* single *Jordan block of size $M{+}1$ at eigenvalue $M\omega$: secular growth t^M, with degree unbounded up the tower.*

Proof. $d\Gamma(N)$ acts on the occupation basis $\mathrm{sym}(u^\nu v^{M-\nu})$ as a weighted shift $\nu \to \nu + 1$ with nonzero weights, so $\big(d\Gamma(N)\big)^M$ maps $\mathrm{sym}(v^M)$ to a nonzero multiple of $\mathrm{sym}(u^M)$ while $\big(d\Gamma(N)\big)^{M+1} = 0$; the kernel of $d\Gamma(N)$ in the sector is one-dimensional (geometric multiplicity 1). □

Verified numerically for $M \le 8$: nilpotent ranks, geometric multiplicities, and fitted growth exponents $1.000, 2.000, 3.999, 5.999, 7.999$ at $M = 1, 2, 4, 6, 8$ (`scripts/t4_sector_tower.py`).

Remark 4.4 (the sector similarity is positive-diagonal). The shift of Proposition 4.3 is not merely nonzero- but *positively*-weighted: on the occupation basis $\mathrm{sym}(u^\nu v^{M-\nu})$ (taking $\{u, v\}$ orthonormal—the convention defining the occupation beables), $d\Gamma(N) = a_u^\dagger a_v$ sends $|\nu\rangle \mapsto \sqrt{(\nu{+}1)(M{-}\nu)}\,|\nu{+}1\rangle$, with strictly positive weights for $\nu = 0, \ldots, M{-}1$. Hence $d\Gamma(N) = D\,J_M\,D^{-1}$ with J_M the standard size-$(M{+}1)$ nilpotent block (ones on the subdiagonal, matching the shift $\nu \to \nu{+}1$) and $D = \mathrm{diag}(d_\nu)$, $d_0 = 1$, $d_{\nu+1} = \sqrt{(\nu{+}1)(M{-}\nu)}\,d_\nu$ ($\nu = 0, \ldots, M{-}1$), a *fixed, positive* diagonal (`scripts/t4_tower_similarity.py`). The occupation and Jordan bases therefore differ, at every M, by one *t-independent* positive-diagonal similarity: $U(t) = e^{-\mathrm{i}M\omega t}\,D\,e^{-\mathrm{i}J_M t}D^{-1}$, so $|U(t)|^2 = D^2\,|e^{-\mathrm{i}J_M t}|^2\,D^{-2}$ entrywise (the global phase $e^{-\mathrm{i}M\omega t}$ cancels), and any two-time object $\Gamma(T)\Gamma(s)^{-1}$ built from $\Gamma = |U|^2$ conjugates by the same D^2, scaling entry (i, j) by $(d_i/d_j)^2 > 0$: at each M the beable-basis divisor shares the entrywise sign pattern of the Jordan-basis one. Moreover $e^{-\mathrm{i}J_M t}$ is triangular *Toeplitz*—its offset-ℓ entry is $(-\mathrm{i}t)^\ell/\ell!$, independent of M—so the Jordan-basis divisor's offset-ℓ entry is one and the same at every $M \ge \ell$; a depth-ℓ sign shown for one block size therefore holds, in the beable basis, at every sector $M \ge \ell$. (t-independence is essential: it is what lets D^2 pass through the two-time divisor $\Gamma(T)\Gamma(s)^{-1}$.)

The hypothesis is Sec. 3's kinematics: A is the one-particle *Hamiltonian* of the equal-frequency pair (real eigenvalue ω), which inherits the nilpotent structure N of the mode matrix's $+\mathrm{i}\omega$ Jordan cell (the positive-frequency one of its two conjugate 2×2 blocks—the eigenvalue is the cell's times i^{-1}, the nilpotent the same up to that scalar, the Jordan structure unchanged; this per-level Jordan structure is the Bender–Mannheim equal-frequency construction [8]), so the Fock tower of every conformal mode carries this structure. This is the constructive form of Smilga's observation that the equal-frequency point is an exceptional point of *infinite order*, whose closure produces a continuous spectrum [15]. Two consequences. First, no truncation escapes the lemma—every level $M \ge 1$ of the tower contains Jordan structure, and each level makes the secular degree worse. Second, the no-representation statement lifts to the field:

Theorem 4.5 (no boundedly invertible dictionary on the Fock tower)**.** *Let $H = d\Gamma(A)$ as above ($A = \omega\mathbf{1} + N$ a size-2 Jordan pair with $\omega \in \mathbb{R}$) on the Fock space of one mode (sector-preserving with finite-dimensional sectors, so $\mathrm{e}^{-\mathrm{i}Ht}$ is defined sector-wise on a dense domain; it is* not *bounded on ℓ^2—its sector norms grow exponentially in M at fixed t), and let S be any bounded operator with bounded inverse on the ℓ^2 configuration space. Then $\Gamma_{ij}(t) = |(S\mathrm{e}^{-\mathrm{i}Ht}S^{-1})_{ij}|^2$ cannot have unit column sums for all t (whenever the representation exists at all, i.e. $\mathrm{e}^{-\mathrm{i}Ht}S^{-1}e_j \in \ell^2$, which we take as part of the dictionary class; an entrywise column that leaves ℓ^2 is not a bounded-in-Fock-norm representation and is excluded by definition).*

Proof. Unit column sums mean $\|V(t)e_j\| = 1$ for every basis vector e_j and all t. Let u, v be the Jordan pair in the one-quantum sector, and let u^* and α be the coordinate functionals dual to u and v there, extended by zero off the sector (bounded); since $\mathrm{e}^{-\mathrm{i}Ht}$ preserves sectors, they read off the one-quantum component, on which H acts as A, giving $\langle u^*, \mathrm{e}^{-\mathrm{i}Ht}\psi\rangle = \mathrm{e}^{-\mathrm{i}\omega t}\big(\langle u^*, \psi\rangle - \mathrm{i}t\,\alpha(\psi)\big)$ for every ψ in the domain. If $\alpha(S^{-1}e_j) = 0$ for every j, then the bounded functional $\alpha \circ S^{-1}$ vanishes on a basis, hence everywhere, hence $\alpha = 0$ (as S^{-1} is surjective)—false. So some j has $\alpha(S^{-1}e_j) = a \neq 0$, and with $c = |\langle u^*, S^{-1}e_j\rangle|$,

$$1 = \|V(t)e_j\| \geq \frac{\big\|\mathrm{e}^{-\mathrm{i}Ht}S^{-1}e_j\big\|}{\|S^{-1}\|} \geq \frac{|a|\,t - c}{\|u^*\|\;\|S^{-1}\|} \longrightarrow \infty. \qquad \square$$

Remark 4.6 (mode-wise extension to the full free Fock space). The theorem extends to the multi-mode free field: $H = \sum_k d\Gamma(A_k)$ commutes with every per-mode number operator $\hat{N}_k$, so the two-dimensional subspace {one quantum in mode k, vacuum elsewhere} is invariant and lies in an invariant orthogonal decomposition; the dual functionals extend by zero and the linear-growth argument of Theorem 4.5 runs verbatim mode-wise.

A control makes the mechanism sharp: for a *diagonalizable* one-particle matrix at the same degenerate frequency, every sector evolution is exactly unitary ($\big|\big\|\mathrm{e}^{-\mathrm{i}d\Gamma t}\big\| - 1\big|$ at machine epsilon, 2.2×10^{-16}, at all sectors and times tested): degeneracy alone is harmless; the Jordan pair is what kills the tower. Scope, stated plainly: this is the *free* field on flat space—precisely the arena whose per-mode kinematics Sec. 3 measured; interactions and curved backgrounds remain outside the theorem. "Bounded" here is relative to the occupation-number ($\{u, v\}$-orthonormal) Fock norm of Remark 4.4: a Dyson map bounded in a different Fock norm falls into the honestly-open unbounded/sector-mixing class of Claim 6.1, not around the theorem.

4.2 What survives: the open process, and the adjudication

What the indivisible-process paper [1] found at the dimer exceptional point holds here: the *open* route is untouched (for the mechanical cell, and sector by sector on the tower—H preserves each finite sector, each carrying its own contraction and dilation; no tower-wide normalization exists, by the unbounded sector norms of Theorem 4.5). Normalizing $C(t) = \mathrm{e}^{-\mathrm{i}Ht}/\sigma_{\max}(t)$, with $\sigma_{\max}$ the largest singular value, gives an exact contraction; its Halmos dilation is unitary (verified to 1.4×10^{-8}, limited by a matrix square root at an exactly-degenerate singular value); and the reduced process is sub-stochastic at all times. The normalization is a convention, and the minimal one: a unitary dilation exists exactly when the normalized evolution is a contraction, so $\sigma_{\max}$ is the pointwise-least normalization admitting a conservative completion—the unique choice that discards no survival probability beyond what the dynamics itself loses ($\|C(t)\| = 1$: the least-decaying state is unconditioned at every time). Its structure is, however, new to the series' taxonomy: for the pure Jordan cell, $|U(t)|^2$ is *triangular*—the beable-level process is a one-way cascade into the growth direction—and its indivisibility enters not through divisor negativity (the broken-dimer mechanism of [1]; the dimer at its own exceptional point develops a negative divisor entry by $t \approx 0.7$ at unit coupling (κ, the cell's off-diagonal coupling)—the EP limit $1/(\kappa\sqrt{2})$ of the companion's onset formula; the companion's canonical $\theta = \pi/4$ device gives $t \approx 1.0$—while the size-2 PU cell never does, its 2×2 divisor being too small to carry a depth-two entry) but through the divisor violating the column-sum condition: *conditional mass creation*, onset at finite length ($t_c = 0.486$ for the unit cell in the zero-step limit—the root of $2t\,\sigma_{\max}^2(t) = (\sigma_{\max}^2)'(t)$, emitted by the script; ≈ 0.50 at this section's protocol, whose base grid

and pair step are both 0.05), with entrywise nonnegativity intact throughout *for the size-2 cell* (whose triangular divisor has nonnegative entries). At the higher Fock sectors $M \geq 2$ the divisor's depth-two entry $\frac{1}{4}(T^2 - s^2)(T^2 - 3s^2)$ turns negative for a short step ($s < T < \sqrt{3}\, s$), carried into the beable basis by the positive-diagonal similarity of Remark 4.4—so there the companion's ordinary negativity mechanism reappears alongside mass creation. One bookkeeping fact must be flagged, because the two mechanisms differ in kind under it: divisor entry *signs* are invariant under every positive scalar renormalization $c(t)$ of the normalization, while divisor *column sums* scale by $c^2(s)/c^2(s+\tau)$—negativity is convention-free, but the mass-creation onset t_c is stated relative to the minimal ($\sigma_{\max}$) convention. The convention is not idle, and its minimality carries the physics: for the unit cell with coupling κ the raw $|U|^2$ divisor's column sum is exactly $1+\kappa^2(T^2-s^2)$, so a renormalization avoids the crossing at all step pairs only if $c^2(t) \geq c^2(0)\, \mathrm{e}^{\kappa^2 t^2}$, and that Gaussian envelope does avoid it (both verified in `scripts/t4_jordan_process.py`, block [3b]). Divisibility can thus be purchased, but only by conditioning away survival probability at a Gaussian rate against polynomially growing dynamics—a conditioning overhead $c^2/\sigma_{\max}^2$ that diverges; under the minimal convention, in which conditioning means survival and nothing else, the crossing is unavoidable. The dipole ghost, read as a process, is a one-way conditioned cascade.

This dissolves the published dispute into two currencies:

- *Smilga's statement* (said of the nonstandard realization: "unitarity is broken in the equal frequencies limit" [14]) is the closed-currency fact, now a lemma: no similarity restores a unitary—hence unistochastic—closed evolution.
- *Bender–Mannheim's construction* (unitary time evolution carried by nonstationary states [8]) is, in this currency, the conditional bookkeeping of the dilated open process (read sector-wise on the tower): the polynomial norm growth is divided out exactly as the broken-phase dimer's exponential growth was in [1]—a survival-conditioned, open-system reading, not a closed sample space.

Both are right where they compute; neither headline survives the translation. We anticipate the natural reply from the $\mathcal{PT}$ side: in the theory's own inner product nothing is conditioned, and probability is exactly conserved. Agreed—and that is precisely the point at issue: the conditioning is relative to the Dirac/beable norm, the only norm that assigns Born weights to *configurations*, so the disagreement is not about the formalism but about whether a configuration sample space exists. That is the trilemma's third seat, not a technicality. And the dictionary line for field theory: the Jordan t-term in amplitudes—the t^2 in probabilities—is precisely the notorious per-mode linear growth of fourth-order-gravity perturbations [41]: conformal gravity is the everywhere-realized member of the exceptional-point polynomial class of [1].

5 Vacuum decay: the contact rung and the α_g fork

Suppose one refuses the exit and keeps definite configurations: the ghost sector is then subject to the vacuum-decay analysis of [2]. There the golden-rule rate factorizes into an infinite boost-orbit volume times an invariant reduced integral, and the reduced measure's ultraviolet scaling is set by the vertex: contact $|\mathcal{M}|^2 \sim 1$ gives Λ^4, the graviton-mediated phantom of Cline–Jeon–Moore, $|\mathcal{M}|^2 \sim s^2/M_{\mathrm{Pl}}^4$, gives $\Lambda^8/M_{\mathrm{Pl}}^4$ [33, 2].

For conformal gravity the physical channel is the Cline–Jeon–Moore analog, vacuum → (ghost-graviton pair) + (photon pair), through the $1/(\alpha_g k^4)$-scaled propagator; the matter side couples through the standard linearized $\frac{1}{2}h_{\mu\nu}T^{\mu\nu}$ coupling (the quadratic-action form of the sourced field equation of [9]). The illustrative T-parallel, with pair matrix elements $\langle \text{pair}|T|0\rangle \sim s$,

$$\mathcal{M} \;\sim\; s \cdot \frac{1}{\alpha_g s^2} \cdot s \;\sim\; \frac{1}{\alpha_g}, \qquad |\mathcal{M}|^2 \sim \frac{1}{\alpha_g^2}\ (s\text{-flat}), \tag{2}$$

carries the correct α_g counting for the physical channel as well (the same α_g^{-1} follows for the ghost-self-interaction route by α_g bookkeeping: each graviton vertex drawn from the $\alpha_g C^2$ action carries α_g, each internal line $1/\alpha_g$, and each canonically normalized external leg $\alpha_g^{-1/2}$, so a connected $0 \to 4$ tree with n_v vertices and $n_v - 1$ internal lines scales as $\alpha_g^{n_v-(n_v-1)-2} = \alpha_g^{-1}$, independent of topology), and the s-flatness is *forced* by scale invariance: pure conformal gravity has no dimensionful parameter, so any tree-level $0 \to 4$ massless amplitude is $s^0 f(\alpha_g; t/s, u/s)$—scale invariance fixes the overall Mandelstam degree, not the angular dependence (assumed integrable; any collinear/soft non-integrability would only raise the rate, consistent with the lower-bound reading). The reduced measure therefore sits on the *contact* rung at the scale-invariant point—with Γ now the vacuum-decay rate (not the Sec. 2 stochastic map) per volume V: $\Gamma/V \sim c_{\rm ps}\,\Lambda^4/\alpha_g^2$—a maximally generous lower bound, since consistently regulating the split poles only raises it (for modes of relative detuning $\ll 1$)—with $c_{\rm ps} = (2\pi)^{-4}$ the phase-space prefactor used throughout (slopes validated against [2] to 10^{-3}; `scripts/t4_rate_ladder.py`). The Planck *mass* has left the suppression: what remains is whatever α_g is. This is the structural sense in which the agravity escape—"cosmologically slow … where ghosts have gravitational interactions only" [32]—does not port to conformal gravity: that estimate is a statement about $M_{\rm Pl}$, and conformal gravity has none in its ghost vertex.

Pricing the Lorentz-violating-cutoff rescue (horn 1 of the trilemma) then requires a value of α_g, and here the theory forks, because no numerical α_g exists in print:

Face-value branch. Matching the flat-space $1/(2\alpha_g k^4)$ exchange to the Mannheim–O'Brien rotation-curve parameters [12] gives $|\alpha_g| \sim 3 \times 10^{109}$ (two normalization conventions differing by a fixed factor 4/3; the in-print sign constraint is negative [11]). At such α_g the energy-density bound on the cutoff is vacuous ($\Lambda_{\rm max} \sim 5 \times 10^7\, M_{\rm Pl}$). But this same face-value flat-space theory is the one shown to fail solar-system tests outright in the weak-field limit (wrong light bending [43], disputed by Mannheim [38]; wrong Newtonian limit [44], disputed in [11]): *the branch that survives the vacuum-decay bound as stated (the contact-rung minimum) is already dead by other observations—or, if Mannheim's rebuttals stand (they invoke the very conformal-source structure that defines the Mannheim branch), one is thereby on the other branch.*

Mannheim branch. Mannheim rejects the face-value inversion (conformal sources; the local Higgs field $S(x)$) [11]; then α_g is unfixed, and for any α_g within eight decades of unity (a conservative figure; the same bound in fact reaches $|\alpha_g| \lesssim 3 \times 10^{12}$, below) the same energy-density bound (method anchored on the phantom vertex, where it reproduces the published gamma-ray bound scale within an order of magnitude: 108 vs 18.8 MeV, the latter being the 95% C.L. limit quoted in [34]; using the critical density $\rho_{\rm cr}$ rather than the $\sim 10^4\times$ smaller radiation background is the *weakest defensible* bound, i.e. maximally generous to the theory) forces

$$\Lambda_{\rm max} = \left(\frac{\alpha_g^2 \rho_{\rm cr}}{t_0\, c_{\rm ps}}\right)^{1/5} \;\sim\; 10^{-8}\,\text{eV} \quad (|\alpha_g| = 1), \tag{3}$$

with t_0 the age of the universe: a ghost sector cut off at a reduced wavelength $\hbar c/\Lambda_{\rm max} \approx 20\,\text{m}$

(physical wavelength ≈130 m)—no laboratory-scale field content at all.

The gap between the branches, priced. The two branches bracket the positions actually taken in print; they do not exhaust the real line, and we close the gap explicitly rather than by silence. Since the bound scales as $\Lambda_{\max} \propto \alpha_g^{2/5}$, laboratory-scale field content ($\Lambda_{\max} \gtrsim$ meV) returns for $|\alpha_g| \gtrsim 3 \times 10^{12}$, and a particle-scale cutoff ($\Lambda_{\max} \sim 100$ MeV) at $|\alpha_g| \sim 10^{40}$: couplings with $|\alpha_g| \gtrsim 3 \times 10^{12}$—the entire unanchored ray above threshold, bounded by nothing from above—would evade the energy bound as stated. That ray is nonetheless closed, and differently on the two branches. On the *face-value* branch the closure is observational and needs no prior: every leading-order (one-exchange) gravitational observable carries a single $1/(2\alpha_g k^4)$ propagator and so scales as $1/\alpha_g$ (nonlinear observables scale as higher powers of $1/\alpha_g$ and close even faster), whence the linear potential coefficient $\gamma_* \propto 1/\alpha_g$ is *measured* by the rotation-curve fit to be $|\alpha_g| \simeq 3 \times 10^{109}$—itself already dead by solar-system tests [43] (a dark-matter-independent, though Mannheim-disputed, anchor)—and any $|\alpha_g|$ off that value fails galactic and solar dynamics monotonically (below it the linear term overshoots the observed pull, above it undershoots by the matching decades). On the *Mannheim* branch α_g decouples—on Mannheim's contention, via degeneracy with the undetermined conformal-source normalization—from every conformally-invariant *classical* observable, so no classical or astrophysical-fit measurement constrains it (graviton–graviton scattering does depend on α_g, but is unobservable here); there the ray is closed by a *naturalness prior*, stated rather than smuggled: a dimensionless coupling is not taken tens of decades above unity without a mechanism. (The Mannheim energy bound already reaches the threshold on its own—$\Lambda_{\max} \lesssim$ meV for all $|\alpha_g| \lesssim 3 \times 10^{12}$—leaving only the ray above it to the prior; and replacing $\rho_{\rm cr}$ by the smaller measured radiation backgrounds tightens $\Lambda_{\max}$ by only the fifth root of the density ratio, a factor of order ten.) Under either branch's closure horn 1 is untenable (on the Mannheim branch up to the stated prior); absent any closure it survives only on the unanchored ray $|\alpha_g| \gtrsim 3 \times 10^{12}$, selected to evade the bound.

Neither branch keeps horn 1 alive, and the unanchored ray of larger couplings is closed on each (observationally on the face-value branch, by the stated prior on Mannheim's). And the escape Mannheim himself offers from both—the $\mathcal{PT}$/zero-norm quantization, under which the decay channel closes identically [7, 10]—is not a fourth option: it is horn 3, and Sec. 4 has already priced it.

Scope. Everything in this section is tree-level, golden-rule, and computed on the Minkowski background with Dirac-norm (Fock) quantization, with the exactly degenerate external pair understood as the smooth $\delta \to 0$ limit of the split-pole spectrum—an assumption, since the merging poles carry opposite residues (the degenerate tower of Prop. 4.3 has no stationary final-state basis at $\delta = 0$)—that is, on the multiplicity (instantaneous-decay-products) sample-space choice of [2], which is exactly the reading in which horn 1 is being priced. Indeed the assumption is not merely unproven but fails in the regulated theory: canonically normalizing the split ($\delta > 0$) external legs against their magnitude-$1/(\alpha_g\delta)$ residues multiplies each gravitational leg by a divergent factor $\propto \delta^{-1/2}$, so the consistently regulated golden-rule rate is parametrically larger than the contact-rung value (for modes of relative detuning $\ll 1$) and has no finite $\delta \to 0$ limit—the split-pole and $\delta = 0$ computations sit on opposite sides of a non-commuting limit. Because these Dirac-norm channel rates are positive and cannot cancel (positivity is norm-relative), this only tightens (never loosens) the horn-1 pricing; the contact rung is the maximally generous (minimum) reading. It is not a statement about the $\mathcal{PT}$-quantized theory, whose channels close by construction; that is the point of the fork, not an oversight of it.

6 The trilemma, closed

Claim 6.1 (conformal gravity must choose—and already has)**.** Of the trilemma's three virtues—a coupled ghost sector, empirical Lorentz invariance, definite configurations—conformal gravity's *no-ghost* kinematics removes the third, within the stated dictionary classes, at the outset: by Lemma 4.1 the equal-frequency (Jordan) dynamics admits no closed configuration-space representation under any dictionary in the stated classes—any similarity per sector, or any dictionary bounded with bounded inverse in the occupation-number Fock norm (Theorem 4.5, extended mode-wise to the full free Fock space by Remark 4.6). (Unbounded, sector-mixing field dictionaries, continuum-beable $(x, \dot{x})$ kernels on L^2, and the deterministic Bohmian configuration trajectories of [19] fall outside these classes and remain open.) Within the stated classes, definite configurations survive only in the Hermitian ghost realization, never jointly with the other two virtues (at tree level). The remaining choices are priced:

- **Horn 1** (coupled ghosts + configurations ⇒ Lorentz-violating frame): the α_g fork of Sec. 5 (priced under the smooth $\delta \to 0$ final-state limit of that section's Scope)—vacuous-but-dead, or a macroscopic-wavelength cutoff. Untenable on either branch (Mannheim up to the stated prior), with the surviving ray of larger couplings closed on each (observationally on the face-value branch, by the naturalness prior on Mannheim's).

- **Horn 2** (Lorentz + configurations ⇒ no coupled ghosts): no conformal gravity coupled to matter—the theory abandons its purpose.

- **Horn 3** (coupled ghosts + Lorentz ⇒ no configurations): the Bender–Mannheim quantization. This is where the theory already lives, by kinematics rather than choice; its price, in this currency, is that the "unitarity" retained is the conditional bookkeeping of an open process (Sec. 4)—at the free/kinematic level, within the stated dictionary classes, there is no underlying *closed* sample space of definite configurations (the lemma and the tower theorem are free-field statements; Sec. 7 scopes them).

The *adjudicated* escape routes of quadratic gravity map onto the horns (two rows only horn-adjacent or premise-level, as flagged in Table 1) with no further remainder we have been able to identify (the recent (2024–2026) entries flagged in the introduction, the older alternative quantizations [31, 42], and the Bohmian route [19], remain pending; Table 1): Lorentz-violating cutoffs and frames [33, 49] (horn 1); Maldacena's boundary conditions [46], argued insufficient on Minkowski [41] (an attempted horn-2 residence); $\mathcal{PT}$/zero-norm quantization [7, 10], fakeons [40], and negative-norm configuration space [35] (horn 3, in three dialects: all exit the sample space, paying respectively in nonstationary-state bookkeeping, microcausality that "survives the classical limit," and probability-free configurations); Merlin modes [39] (horn 3 adjacent: the sample space is kept but causal order is not, which in the operational terms of [2] is a different door out of the same room); and the trade of Weyl ghosts for observable Lorentz violation in the primordial gravitational-wave spectrum [47] is a horn-1 special case (with a horn-2 aspect) computed before the general pattern was visible. The closest prior statements of a forced choice—Donoghue–Menezes' unitarity at the price of microcausality (violated on scales of order the resonance width) [39] and CJM's stability-versus-Lorentz [33]—each name one edge, and Steinwachs names a different triangle altogether (unitarity–renormalizability–relativistic invariance) [48]; the present triangle, and the identification of its third vertex with the existence of a configuration sample space, appears to be new here, as does every use of the process currency in this corpus.

escape route	horn	price, in the process currency
Lorentz-violating cutoff / preferred frame [33, 49]	1	the α_g fork: vacuous-but-observationally-dead, or a macroscopic-wavelength cutoff (larger-coupling ray closed by fit or stated prior)
Lorentz-violating gravitons [47]	1	observable blow-up of the primordial GW spectrum (their result; a two-horn special case)
decouple the ghost sector	2	no conformal gravity coupled to matter
Maldacena boundary conditions [46]	2 (attempted)	argued insufficient on Minkowski [41]
$\mathcal{PT}$/zero-norm quantization [7, 10]	3	no *closed* sample space within the stated dictionary classes (Lemma 4.1); retained unitarity is conditioned, open-system bookkeeping
negative-norm configuration space [35]	3	configurations without a probability measure
fakeons [40]	3	definition-level exit; microcausality violation "survives the classical limit"
Merlin modes [39]	3-adjacent	sample space kept, causal order paid (a loop-level width construction, outside the tree-level trilemma)
metastability / benign ghosts [13, 32, 37, 27, 28, 29]	(premise)	disputes the rate premise rather than picking a horn; the open QFT rate is conceded "naively infinite"; the 2026 quantum-mechanical entries sharpen the dispute (unitary evolution with a well-defined vacuum; exact moment bounds) without touching the field-theoretic rate

Table 1: The adjudicated published escape routes of quadratic/conformal gravity (one cited work concerns massive gravity's Boulware–Deser ghost [49]), assigned to the trilemma's horns. No route occupies a fourth position that we can identify; the closest candidate is the Merlin route, which keeps the sample space and trades causal order instead—a different door out of the same room, priced in a currency the trilemma does not list but operational physics does.

7 Discussion: what is established, at what strength

Theorem-grade: Lemma 4.1 and its corollary that the conformal point's Jordan (no-ghost) dynamics admits no closed beable representation (per sector; on the Fock tower for dictionaries boundedly invertible in the occupation-number Fock norm); the sector-tower proposition and the bounded-dictionary theorem on the free Fock space (Sec. 4.1); the validity of the open (dilated) process for the mechanical Jordan cell (Sec. 4). *Measured*: the reduced-measure slopes (validated against [2]); the per-mode dictionary cost (1); the UV approach to the exceptional line with its units-honest decomposition; the conditional-mass-creation indivisibility mechanism of the dipole cell (under the minimal $\sigma_{\max}$ convention, Sec. 4). *Argued*: the adjudication of Sec. 4 (that Bender–Mannheim's nonstationary-state unitarity *is* the dilation's conditional dynamics is an identification we find exact in structure, but it is an interpretation of their formalism, not a theorem about it); the one-loop pole statement of Sec. 3 that the Merlin width does not detune the theory off the exceptional line (a resonance-type EP with unequal linewidths—an analytic one-loop estimate); the $T \sim s$ vertex parallel behind the contact rung, and with it the reduced-measure power counting (its s-flatness is forced by scale invariance and its slopes are measured, but the vertex class itself is the argued premise); the horn assignments of Sec. 6. *Conditional*: everything in Sec. 5, on the stated branch structure—in particular the $|\alpha_g| \sim 3 \times 10^{109}$ face-value

matching is our arithmetic on published fit parameters, not a published number, and Mannheim disputes the matching's premise; we present the fork, not a verdict on it.

Limits. All rate statements are tree-level golden rule on flat space with Dirac-norm quantization; Mannheim contends flat space with those norms is the wrong arena twice over, and our reply is the structure of the paper: on his arena the trilemma's third seat is already gone (Sec. 4), so the rate section prices only the alternative he rejects. The energy-density bound is deliberately crude (its calibration against the phantom case is order-of-magnitude); a dedicated soft-photon analysis would tighten but not rescue horn 1. The equal-frequency quantum mechanics we adjudicate is no longer only the mechanical prototype—the sector tower (Sec. 4.1) carries the no-representation statement to the free Fock space of every conformal mode, for any dictionary boundedly invertible in the occupation-number Fock norm—but the field-theoretic zero-norm construction still has structure (conformal ghosts on curved backgrounds, the $S(x)$ sector, interactions) that the free flat-space tower does not see, and our theorems constrain the latter only. Finally, the trilemma names combinations, not truths: conformal gravity may yet be the right theory of nature—the claim is that if so, it is a theory without a configuration sample space, and its advertised unitarity is of the conditioned, open-system kind.

Author contributions and use of AI

This manuscript was produced through a collaboration, directed by the author, with the large language model Claude Fable 5 (Anthropic), which carried out the technical work: the research program, the derivations and proofs, the numerical experiments, the literature analysis, and the text, including this statement. J.T. posed the research direction, provided iterative review, and takes responsibility for the decision to circulate. The research notebook in the program's public repository records the development, including a same-day correction of this paper's own initial cutoff-bound claim (superseded by the α_g fork of Sec. 5) when the coupling's face value was pinned down.

Code availability

All numerical claims are reproducible from five standalone scripts (NumPy/SciPy, seconds to minutes, controls printed inline) in the program's main repository, with companion-paper numbers cited as such, the face-value α_g matching emitted by `scripts/t4_rate_ladder.py` from the constants documented in `research/t4_vertex_conventions.md`, and the one-loop pole estimates of Sec. 3 (width, suppression factor, fixed-linewidth foil) analytic computations whose arithmetic is stated in-text:

https://github.com/embeddetech/p05-choosing-ghosts

(also archived, with the full program history, in the hub repository https://github.com/embeddetech/ghosts-program); `scripts/t4_ep_dictionary.py` (Sec. 3), `scripts/t4_jordan_process.py` (Sec. 4), `scripts/t4_sector_tower.py` and `scripts/t4_tower_similarity.py` (Sec. 4.1), `scripts/t4_rate_ladder.py` (Sec. 5); the supporting literature and convention reports are under `research/`.

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